

Problem 1: Student Grade Calculator

Problem Statement

Write a Java program to accept the marks of five subjects (out of 100 each) using `BufferedReader`.

Calculate:

🎬 Total Marks

🎬 Percentage

🎬 Grade according to the following criteria:

Percentage Grade

≥ 90 A+

80–89 A

70–79 B

60–69 C

50–59 D

< 50 Fail

Also display whether the student has passed or failed.

Concepts Covered

🎬 Variables

🎬 Data Types

🎬 Arithmetic Operators

🎬 if-else ladder

🎬 `BufferedReader`

🎬 Type Casting

Solution--->

https://github.com/ArpitVector/Java_Lab_Assignments/blob/main/Ques1.java

Problem 2: Electricity Bill Generator

Problem Statement

Write a Java program that calculates the electricity bill based on the following slab rates:

Units Rate (₹/Unit)

First 100 units ₹3

Next 100 units ₹5

Next 300 units ₹7

Above 500 units ₹10

Accept:

🎬 Consumer Name

🎬 Consumer Number

🎬 Units Consumed

Display:

🎬 Consumer Details

🎬 Total Bill Amount

If the bill exceeds ₹5000, apply a 5% surcharge.

Concepts Covered

🎬 Nested if-else

🎬 Arithmetic Operators

🎬 BufferedReader

🎬 Variables

Solution-->

https://github.com/ArpitVector/Java_Lab_Assignments/blob/main/Ques2.java

Problem 3: Number Analysis Tool

Problem Statement

Write a menu-driven Java program that accepts an integer and performs the following operations:

1. Check whether the number is Prime
2. Check whether it is Palindrome
3. Check whether it is Armstrong
4. Find the Sum of Digits
5. Reverse the Number
6. Exit

The program should continue until the user selects Exit.

Concepts Covered

- 🎬 switch-case
- 🎬 while loop
- 🎬 do-while loop
- 🎬 Methods
- 🎬 Modulus Operator

Solution--->

https://github.com/ArpitVector/Java_Lab_Assignments/blob/main/Ques3.java

Problem 4: Array Statistics

Problem Statement

Write a Java program to read 10 integers into an array and perform the following operations:

- 🎬 Find Maximum Element
- 🎬 Find Minimum Element
- 🎬 Calculate Average
- 🎬 Count Even Numbers
- 🎬 Count Odd Numbers
- 🎬 Search for a given number
- 🎬 Display the array in reverse order

Concepts Covered

- 🎬 Arrays
- 🎬 Loops
- 🎬 if statement
- 🎬 Searching
- 🎬 Variables

Solution--->

https://github.com/ArpitVector/Java_Lab_Assignments/blob/main/Ques4.java

Problem 5: Employee Salary Management System

Problem Statement

Write a Java program that accepts details of N employees (using arrays):

🎬 Employee ID

🎬 Employee Name

🎬 Basic Salary

Calculate:

🎬 $DA = 40\% \text{ of Basic Salary}$

🎬 $HRA = 20\% \text{ of Basic Salary}$

🎬 $PF = 12\% \text{ of Basic Salary}$

🎬 $Gross\ Salary = Basic + DA + HRA$

🎬 $Net\ Salary = Gross\ Salary - PF$

Display all employee details in a tabular format and also display:

🎬 Highest Paid Employee

🎬 Lowest Paid Employee

🎬 Average Salary of all employees

Concepts Covered

🎬 Arrays

🎬 Loops

🎬 BufferedReader

🎬 Arithmetic Operations

🎬 String Handling

🎬 Methods

Solution--->

https://github.com/ArpitVector/Java_Lab_Assignments/blob/main/Ques5.java